

Plectis

Research with AI, mathematics and software

WILL COOK · OCTOBER 2026

I built Plectis to do research I find interesting with AI, in a way where people who actually know the field can steer it. Basically, the idea is that human mathematicians provide the direction: I prepare results for someone who can interpret them, their feedback changes what I work on next, and I come back with that. I've struggled to get that direction, so a lot of this is me seeing how far you can get without it. When someone has told me what was wrong it changed the work, and the [credit ledger](#)¹ records each case and what it changed, with names withheld until each person says they're happy to be named.

I'm clearly not a mathematician. I chose maths because a proof can be checked, and I realised early on that formal verification is necessary but not sufficient for a lay person using AI to do maths. It's necessary because I can't check a proof myself, so Lean is how I know a statement is actually proved. It isn't sufficient because Lean won't tell you whether the statement is the one you meant, whether it was worth asking, or whether anyone can follow why it's true. The name comes from [plectere](#)², to weave or twist, bringing different capabilities into the same work.

A lot of mathematicians wrote about exactly this in September. [Grant Sanderson](#)³ says every AI-generated proof is born an unsolved exposition problem, and that there will never be Lean for motivated explanations. [De Toffoli and Duede](#)⁴ separate a logical proof, which Lean can certify, from an intelligible one, and say OpenAI's Lean-checked Navier–Stokes result hasn't clearly given mathematicians the second kind yet. [Henry Cohn](#)⁵ calls unexplained AI proofs technical debt that somebody else has to pay, but he's also delighted when non-mathematicians document AI-assisted work responsibly, and asks anyone unsure of their own result to talk to people in the field and shed light on how it was found. That's roughly the standard I'm trying to meet, intelligible arguments, appropriate checks, clear limitations and attribution. Meeting it earns nobody a review, I know that, but if experts favour well-prepared work then whoever produces it has a reason to cut the burden they impose.

The [mathematics repository](#)⁶ is my attempt at that, around eight deliberately difficult Erdős problems. The strongest results are each checked in Lean, and most also have a Comparator check. At [#257](#), the sum of $1/(b^a - 1)$ over any infinite set of finite prime-weighted mass is irrational at every integer base $b \geq 2$, which covers every set whose reciprocals have a finite sum. At [#1041](#), Lean confirms ani's degree-seven polynomial as a counterexample to the Formal Conjectures statement, though whether that matches the 1958 wording still needs independent review. There are partial results at [#243](#), [#269](#) and [#1049](#), the last specialising Zudilin's 2004 construction with no claim to be new. The other seven questions remain open.

What I think is my main contribution is the pipeline that distils it into something people can read, so Lean for the logical part, a short paper for the intelligible part, a long paper for every route that failed and how hard each nearby question turned out to be, which is close to the careful analysis of the difficulty landscape [Terence Tao](#)⁷ asks for, and seminar-style videos we're trying to automate, because as [Daniel Litt](#)⁸ says, talks now show understanding better than papers do. It's more of a proof of concept than me claiming I understand the mathematics deeply enough to explain it. If someone unknown with no understanding can get even part of the way, I think that says something about what an expert could do with the same tools.

Nineteenth-century mariners [contributed observations and received improved sailing charts](#)⁹, so each voyage built on the last. Plectis keeps failed routes beside successful lemmas, with their conditions and limits, for someone else or a future model to use. The [website](#) is the human wrapper. Its [mathematics map](#) connects questions, claims, papers and formal source, the [glossary](#) explains the terms, and a downloadable [source brief](#) lets you explore with your own assistant. If AI is any level more intelligent than a person, I see no reason why that intelligence can't be applied to turning the mathematics it produces back into something people can interpret.

The [software repository](#)¹⁰ applies the same idea beyond mathematics: as AI raises what a back end can do, the front end has to rise with it, so that understanding and auditability keep up with capability. I [published testable parts](#) because I wasn't sure about releasing the whole system. The [safety doctrine](#) sets out the rules my agents work under, my [June questions](#) ask how capability and independent scrutiny interact, and the [systems](#) and [archived open-source](#) papers explain how people can contribute direction, review, exposition, software or compute. My aim is for improving AI to multiply what an expert can direct and understand. If you solve a result, I'll ask for zero priority and it's yours.

SOURCES ¹Plectis credit ledger, [github.com/wcook04/plectis-erdos/.../CREDIT_LEDGER.md](#). ²Corpus of Latin Dictionaries (BBAW), [plecto.cld.bbaw.de/lemma/lat/plecto](#). ³G. Sanderson, *If math is more than proof, we need to better celebrate the rest of it*, 18 September 2026, [terrytao.wordpress.com/2026/09/18](#). ⁴S. De Toffoli and E. Duede, *After Math*, 12 September 2026, [terrytao.wordpress.com/2026/09/12/after-math](#). ⁵H. Cohn, *The technical debt of AI-generated mathematics*, 15 September 2026, [terrytao.wordpress.com/2026/09/15](#). ⁶Plectis mathematics repository, [github.com/wcook04/plectis-erdos](#). ⁷T. Tao, on mining open problems, Mathstodon, 8 September 2026, [mathstodon.xyz/@tao/117237320796901560](#). ⁸D. Litt, *A beginning for mathematics*, September 2026, [daniellitt.com/blog](#). ⁹*The Origin and Work of the Division of Marine Meteorology*, U.S. Naval Institute Proceedings, 1893, [usni.org/magazines/proceedings/1893/july](#). ¹⁰Plectis software repository, [github.com/wcook04/plectis](#).